

Exploratory Data Analysis Plots

Figure 1: Sentiment Distribution Across Review Length Categories

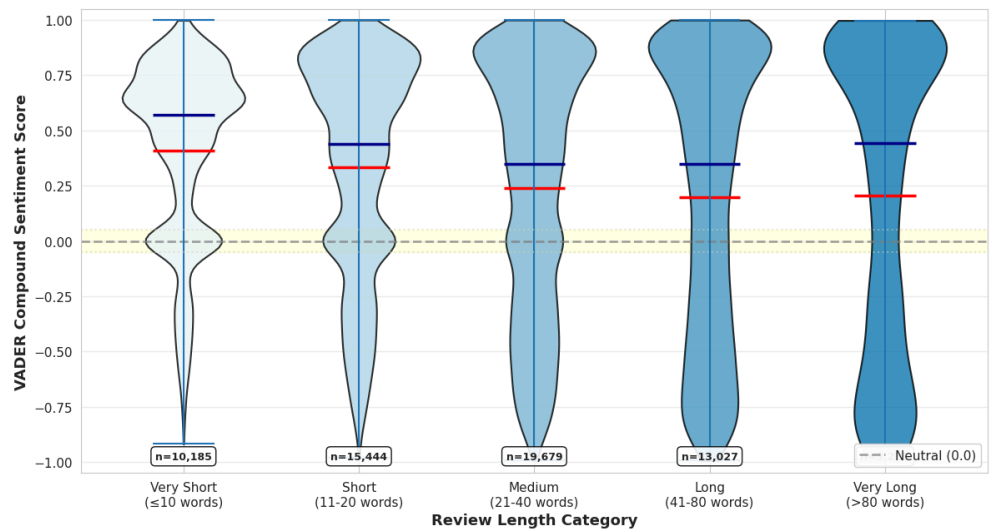
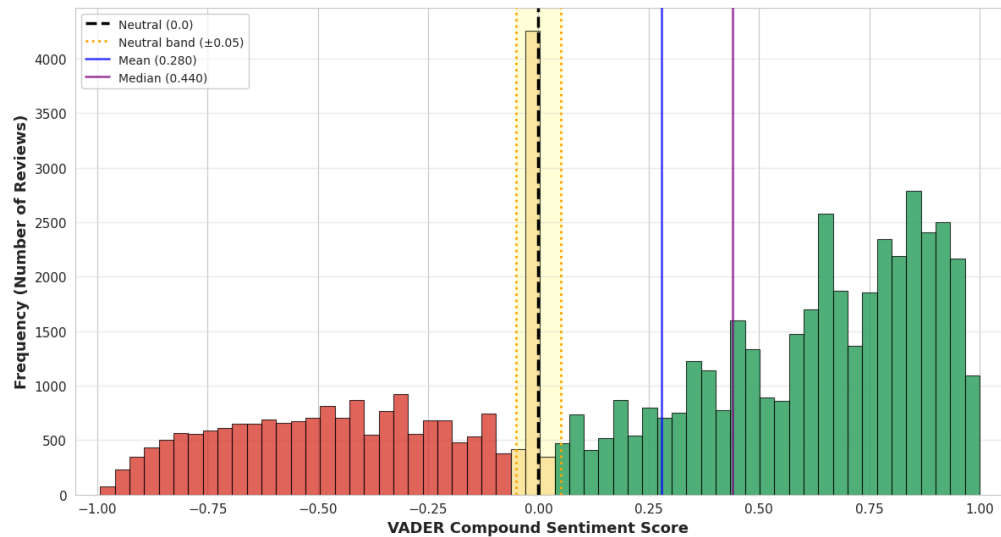
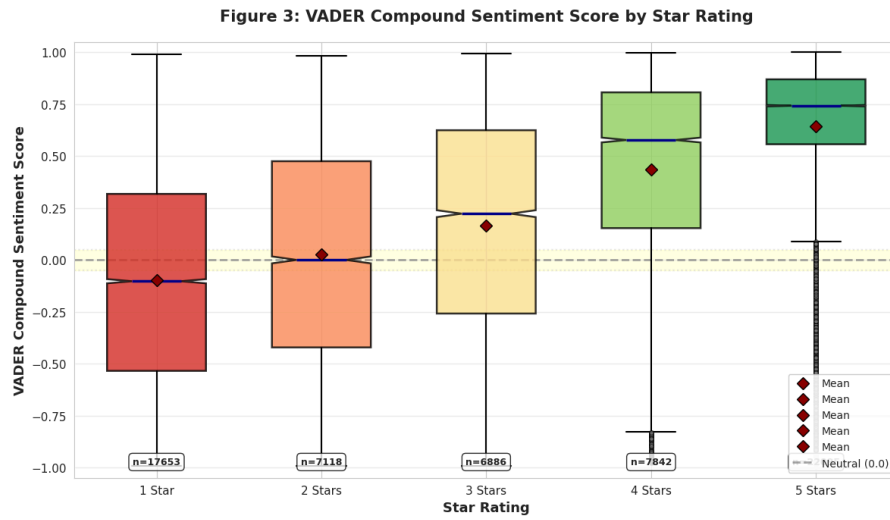
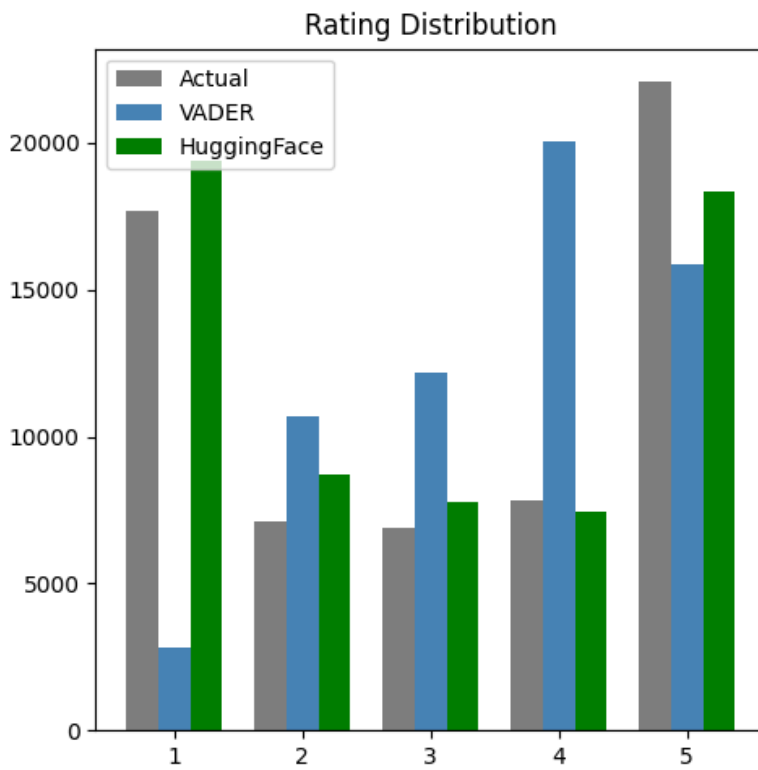


Figure 2: Distribution of VADER Compound Sentiment Scores





Rating Distribution



Output with Emojis

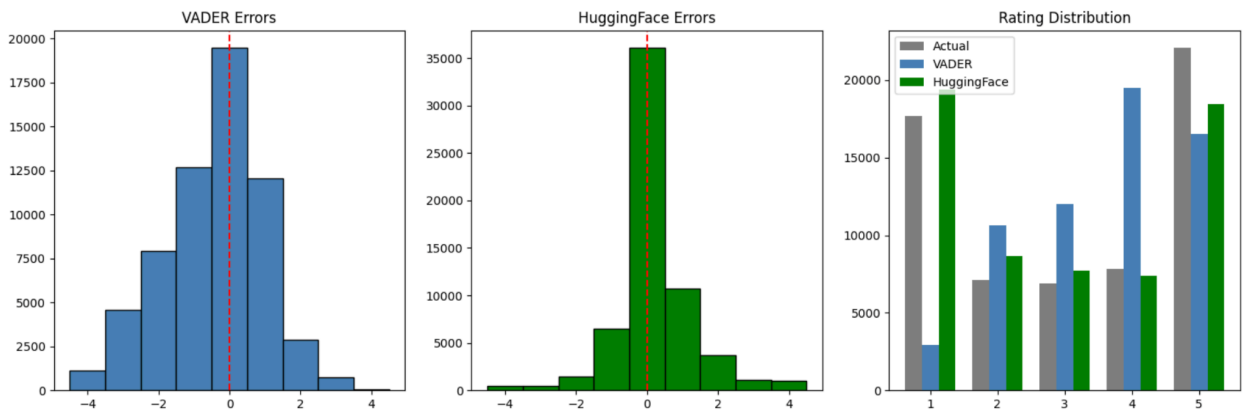
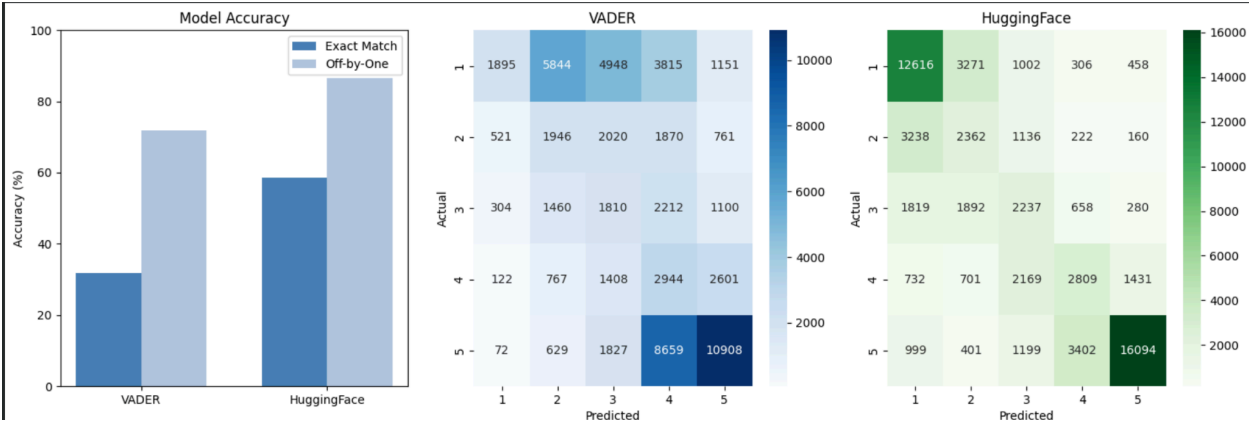
	Review	Rating	vader_compound	vader_predicted_rating
0	Great music service, the audio is high quality...	5	0.9211	5
1	Please ignore previous negative rating. This a...	5	0.6249	4
2	This pop-up "Get the best Spotify experience o...	4	0.5859	4
3	Really buggy and terrible to use as of recently	1	-0.5209	2
4	Dear Spotify why do I get songs that I didn't ...	1	0.7149	4
...	...	...	...	...
61589	Even though it was communicated that lyrics fe...	1	-0.2960	2
61590	Use to be sooo good back when I had it, and wh...	1	0.9143	5
61591	This app would be good if not for it taking ov...	2	-0.9633	1
61592	The app is good hard to navigate and won't jus...	2	0.8074	5
61593	Its good but sometimes it doesnt load the musi...	4	0.8641	5

61594 rows x 4 columns

	Review	Rating	vader_compound	vader_predicted_rating	hf_predicted_rating
0	Great music service, the audio is high quality...	5	0.9211	5	5
1	Please ignore previous negative rating. This a...	5	0.6249	4	5
2	This pop-up "Get the best Spotify experience o...	4	0.5859	4	1
3	Really buggy and terrible to use as of recently	1	-0.5209	2	1
4	Dear Spotify why do I get songs that I didn't ...	1	0.7149	4	1
...	...	...	...	...	...
61589	Even though it was communicated that lyrics fe...	1	-0.2960	2	1
61590	Use to be sooo good back when I had it, and wh...	1	0.9143	5	1
61591	This app would be good if not for it taking ov...	2	-0.9633	1	2
61592	The app is good hard to navigate and won't jus...	2	0.8074	5	3
61593	Its good but sometimes it doesnt load the musi...	4	0.8641	5	3

61594 rows x 5 columns

	Model	Exact Match %	Off-by-One %	Avg Error
0	VADER	31.663798	71.805695	1.091502
1	HuggingFace	58.638828	86.558756	0.621294



	Review	Rating	vader_predicted_rating	hf_predicted_rating	vader_error	hf_error
1	Please ignore previous negative rating. This a...	5	4	5	1	0
3	Really buggy and terrible to use as of recently	1	2	1	1	0
4	Dear Spotify why do I get songs that I didn't ...	1	4	1	3	0
6	I love the selection and the lyrics are provid...	5	4	5	1	0
9	I'm deleting this app, for the following reaso...	1	4	1	3	0

	Review	Rating	vader_predicted_rating	hf_predicted_rating	vader_error	hf_error
2	This pop-up "Get the best Spotify experience o...	4	4	1	0	3
8	It's a great app and the best mp3 music app I ...	5	5	4	0	1
21	His voice is so sweet and hearable	4	4	5	0	1
22	I logged out on my acc but when I tried to ope...	3	3	1	0	2
24	An amazing music experience	4	4	5	0	1

Review	
0	Great music service, the audio is high quality and the app is easy to use. Also very quick and friendly support.
1	Please ignore previous negative rating. This app is super great. I give it five stars+
2	This pop-up "Get the best Spotify experience on Android 12" is too annoying. Please let's get rid of this.
3	Really buggy and terrible to use as of recently
4	Dear Spotify why do I get songs that I didn't put on my playlist??? And why do we have shuffle play?
5	The player controls sometimes disappear for no reason. App restart forgets what I was playing but fixes the issue.
6	I love the selection and the lyrics are provided with the song you're listening to!
7	Still extremely slow when changing storage to external sd card.. I'm convinced this is done on purpose, spotify knows of this issue and has done NOTHING to solve it! Over time I have changed sd cards, each being faster in read, write speeds(all samsung brand). And please add "don't like song" so it will never appear again in my searches or playlists.
8	It's a great app and the best mp3 music app I have ever used but there is one problem that, why can't we play some songs or find some songs? despite this the app is wonderful I recommend it. it's just the best.
9	I'm deleting this app, for the following reasons: This app now has a failing business model. Whether streaming services like it, or not: the consumer doesn't want to pay for music they can't fully own, and 6 ads successively, upon logging in, before a single song, is too much. Closed the app during ad number 6, and I'm more patient than most. If those are the only ways you can profit: you've already peaked. All that's left is your decline.
10	Love Spotify, and usually this app is the best, but as others have stated; the control buttons disappear and podcasts stop for no reason, which is only fixed by deleting and reinstalling the app. First world problems for sure, but highly annoying nonetheless, especially when paying for prime.
11	Can't play Spotify when on WiFi
12	I had amazon premium music family package and it was good. Everyone could listen to whatever they liked on their respective alexas in their rooms by just asking to play, no problem. We now have spotify premium family and the alexa integration is poor. It just doesn't like spotify. It just stops playing all of the time , you can't play on different alexa devices at the same time just by asking. Why isn't the integration better?
13	Worst app always says I'm offline and never showed any song or podcast
14	I hav any music that I like it is super👍👍

Output Without Emojis

	Review	Rating	Review_cleaned
0	Great music service, the audio is high quality...	5	Great music service, the audio is high quality...
1	Please ignore previous negative rating. This a...	5	Please ignore previous negative rating. This a...
2	This pop-up "Get the best Spotify experience o...	4	This pop-up "Get the best Spotify experience o...
3	Really buggy and terrible to use as of recently	1	Really buggy and terrible to use as of recently
4	Dear Spotify why do I get songs that I didn't ...	1	Dear Spotify why do I get songs that I didn't ...
...	...	...	...
61589	Even though it was communicated that lyrics fe...	1	Even though it was communicated that lyrics fe...
61590	Use to be sooo good back when I had it, and wh...	1	Use to be sooo good back when I had it, and wh...
61591	This app would be good if not for it taking ov...	2	This app would be good if not for it taking ov...
61592	The app is good hard to navigate and won't jus...	2	The app is good hard to navigate and won't jus...
61593	Its good but sometimes it doesnt load the musi...	4	Its good but sometimes it doesnt load the musi...
61594 rows × 3 columns			

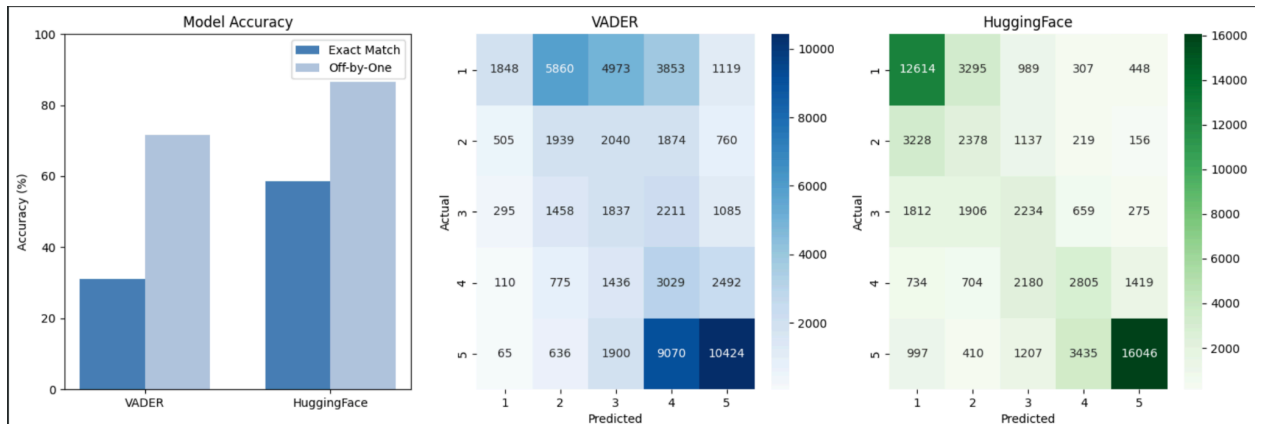
	Review	Rating	Review_cleaned	vader_compound	vader_predicted_rating
0	Great music service, the audio is high quality...	5	Great music service, the audio is high quality...	0.9211	5
1	Please ignore previous negative rating. This a...	5	Please ignore previous negative rating. This a...	0.6249	4
2	This pop-up "Get the best Spotify experience o...	4	This pop-up "Get the best Spotify experience o...	0.5859	4
3	Really buggy and terrible to use as of recently	1	Really buggy and terrible to use as of recently	-0.5209	2
4	Dear Spotify why do I get songs that I didn't ...	1	Dear Spotify why do I get songs that I didn't ...	0.7149	4
...	...	...	...	...	...
61589	Even though it was communicated that lyrics fe...	1	Even though it was communicated that lyrics fe...	-0.2960	2
61590	Use to be sooo good back when I had it, and wh...	1	Use to be sooo good back when I had it, and wh...	0.9143	5
61591	This app would be good if not for it taking ov...	2	This app would be good if not for it taking ov...	-0.9633	1
61592	The app is good hard to navigate and won't jus...	2	The app is good hard to navigate and won't jus...	0.8074	5
61593	Its good but sometimes it doesnt load the musi...	4	Its good but sometimes it doesnt load the musi...	0.8641	5

61594 rows x 5 columns

	Review	Rating	Review_cleaned	vader_compound	vader_predicted_rating	hf_predicted_rating
0	Great music service, the audio is high quality...	5	Great music service, the audio is high quality...	0.9211	5	5
1	Please ignore previous negative rating. This a...	5	Please ignore previous negative rating. This a...	0.6249	4	5
2	This pop-up "Get the best Spotify experience o...	4	This pop-up "Get the best Spotify experience o...	0.5859	4	1
3	Really buggy and terrible to use as of recently	1	Really buggy and terrible to use as of recently	-0.5209	2	1
4	Dear Spotify why do I get songs that I didn't ...	1	Dear Spotify why do I get songs that I didn't ...	0.7149	4	1
...	...	...	...	...	...	...
61589	Even though it was communicated that lyrics fe...	1	Even though it was communicated that lyrics fe...	-0.2960	2	1
61590	Use to be sooo good back when I had it, and wh...	1	Use to be sooo good back when I had it, and wh...	0.9143	5	1
61591	This app would be good if not for it taking ov...	2	This app would be good if not for it taking ov...	-0.9633	1	2
61592	The app is good hard to navigate and won't jus...	2	The app is good hard to navigate and won't jus...	0.8074	5	3
61593	Its good but sometimes it doesnt load the musi...	4	Its good but sometimes it doesnt load the musi...	0.8641	5	3

61594 rows x 6 columns

	Model	Exact Match %	Off-by-One %	Avg Error
0	VADER	30.972173	71.677436	1.098954
1	HuggingFace	58.572264	86.592850	0.621359



	Review	Rating	vader_predicted_rating	hf_predicted_rating	vader_error	hf_error
1	Please ignore previous negative rating. This a...	5	4	5	1	0
3	Really buggy and terrible to use as of recently	1	2	1	1	0
4	Dear Spotify why do I get songs that I didn't ...	1	4	1	3	0
6	I love the selection and the lyrics are provid...	5	4	5	1	0
9	I'm deleting this app, for the following reaso...	1	4	1	3	0

	Review	Rating	vader_predicted_rating	hf_predicted_rating	vader_error	hf_error
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8	It's a great app and the best mp3 music app I ...	5	5	4	0	1
21	His voice is so sweet and hearable	4	4	5	0	1
22	I logged out on my acc but when I tried to ope...	3	3	1	0	2
24	An amazing music experience	4	4	5	0	1